Assessed Project Part 1 – Group Supervision Week 3

Theme: Designing a sound small-data experiment

The central project question

- Central question: *How does limited training data affect the performance, stability and reliability of classifiers?*

The goal is not simply to find the "best classifier" and compare accuracies, but to design a fair experiment to understand what changes as data become scarce.

Question 1: How would you design a fair experiment to study the effect of training-set size?

- several progressively smaller training sizes;
- same classifiers and metrics across sizes;
- same preprocessing/evaluation procedure;
- stratification where appropriate;
- no test-set leakage.

Key message: *Vary training size; control everything else as much as possible.*

Question 2: If one small training set gives 90% accuracy, how much confidence should you have in that result?

- one split may be lucky/unlucky;
- repeat experiments;
- use repeated stratified splits or cross-validation;
- report mean performance **and variability**;
- learning curves with error bars/bands are useful.

Key message: *With small data, stability matters as much as average performance.*

Question 3: What evidence would show that small data are causing problems?

- lower test/validation performance;
- greater variation across runs;
- widening train–test gap;
- unstable model rankings;
- strong sensitivity to model complexity or the particular split.

Key message: *Small data can make conclusions unreliable, not just reduce accuracy.*